

4th SEMESTER

S. No.	Course Code	Course	Contact Hours per Week (L T P)	Mid Term	End Term	Total Marks	Credits
1	PCC 105	Heat Transfer	3 1 –	50	50	100	4
2	ESC 105	Engineering Materials	3 1 –	50	50	100	4
3	PCC 111	Chemical Technology (Organic)	3 – –	50	50	100	3
4	ESC 104	Strength of Materials	3 1 –	50	50	100	4
5	MBA 105	Production & Operations Management	4 – –	50	50	100	4
6	MBA 106	Managerial Economics	4 – –	50	50	100	4
7	PCC 153	Heat Transfer Lab.	– – 3	50	–	50	1.5
8	PCC 156	Chemical Technology (Organic) Lab.	– – 3	50	–	50	1.5
9	CHE 101	Comprehensive Viva	– – –	50	–	50	1
10	ESC 155	Process Equipment Design	– – 3	50	–	50	1.5

Course Title: HEAT TRANSFER

Course Code: PCC105

Unit I – Conduction

- Steady-state conduction in one-dimensional systems
 - General conduction equation
 - Effect of variable thermal conductivity
 - Steady-state conduction with internal heat generation
 - Lagging on pipes
 - Critical thickness of insulation on pipes
 - Extended surfaces of uniform thickness
 - Fin effectiveness and fin efficiency
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Unit II – Convection and Radiation

Convection

- Free and forced convection
- Concept of heat transfer coefficient
- Dimensionless numbers in free and forced convection
- Dimensional analysis
- Experimental determination of heat transfer coefficient
- Common working correlations

Radiation Heat Transfer

- Black body radiation and gray body radiation
 - Physical mechanism of radiation
 - Radiation properties and shape factor
 - Heat exchange between non-black bodies
 - Radiation shields
 - Pyrometry and effect of radiation on temperature measurement
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Unit III – Condensation, Boiling, and Evaporation

Condensation and Boiling

- Condensation heat transfer phenomenon
- Film condensation on vertical plates and cylinders
- Condensation on horizontal cylinders
- Effect of non-condensable gases and vapor velocity
- Pool boiling and forced convection boiling
- Working correlations for pool boiling

Evaporation

- Types of evaporators
 - Single-effect and multiple-effect evaporators
 - Single and multiple-effect calculations
 - Evaporator capacity and economy
 - Effect of liquid head and boiling point elevation
 - Methods of feeding
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Unit IV – Heat Exchangers

- Types of heat exchangers
- Overall heat transfer coefficient
- Mean temperature difference
- Heat exchanger effectiveness

- Number of transfer units (NTU method)
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Text Books

1. Holman, J.P., *Basic Principles and Calculations in Chemical Engineering*
 2. McCabe, W.L., Smith, J.C., *Unit Operations of Chemical Engineering*, McGraw-Hill
 3. Kern, D.Q., *Process Heat Transfer*, McGraw-Hill
 4. Geankoplis, C.J., *Transport Processes and Unit Operations*, PHI, 3rd Edition, 1999
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Reference Books

1. McAdams, W.H., *Heat Transmission*, McGraw-Hill
 2. Chapman, A.J., *Heat Transfer*, McMillan Publishing
 3. Kreith, F., *Principles of Heat Transfer*, Harper & Row, London
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Course Title: ENGINEERING MATERIALS

Course Code: ESC 105

Unit I – Crystal Structure

- Space lattice and crystal systems
- Close-packed morphology:
 - Hexagonal close packing (HCP)
 - Cubic close packing (CCP)
- Interstitial spaces
- Miller indices
- Linear and planar densities in crystals
- Single crystal and polycrystalline structures
- Structure of ceramics:
 - NaCl structure
 - Zinc blende
 - CsCl
 - Silica and silicates
 - Diamond crystal
- Effect of radius ratio on coordination number

Unit II – Imperfections and Diffusion

- Imperfections in atomic arrangement:
 - Point defects
 - Line defects
 - Surface defects
- Diffusion phenomenon in solids
- Fick's First Law and Second Law of diffusion
- Solid solutions
- Slip systems
- Methods of strengthening materials:
 - Grain size reduction
 - Solid solution strengthening
 - Strain hardening
- Schmid's law

Unit III – Phase Diagrams and Phase Transformation

- Binary phase diagrams:
 - Iron–Iron carbide (Fe–Fe₃C)
 - Copper–Nickel (Cu–Ni)
 - Lead–Tin (Pb–Sn)
- Microstructure development
- Time–Temperature–Transformation (TTT) diagrams
- Heat treatment processes:
 - Hot working
 - Cold working
- Hardening and softening processes

Unit IV – Engineering Materials, Mechanical Testing, and Corrosion

Engineering Materials

- Standards and specifications
- Unified Alloy Numbering System (UNS)
- Ferrous metals and alloys
- Non-ferrous metals and alloys
- Overview of:
 - Ceramic materials
 - Polymeric materials
 - Composite materials

Mechanical Testing

- Standard test procedures for determining:
 - Strength
 - Toughness
 - Fracture toughness
 - Hardness
 - Impact resistance
 - Fatigue behavior
 - Creep behavior

Corrosion

- Types and mechanisms of corrosion
 - Factors influencing corrosion
 - Methods of combating corrosion
 - Selection of materials of construction for handling:
 - Sulfuric acid
 - Nitric acid
 - Sodium hydroxide (NaOH)
 - Hydrochloric acid (HCl)
 - Acetic acid
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Course Title: CHEMICAL TECHNOLOGY (ORGANIC)

Course Code: PCC 11

UNIT I – Oils, Fats, Soaps and Detergents

Oils and Fats

- Introduction
- Extraction of oils from vegetable sources
- Refining of oils and fats
- Hydrogenation of oils

Soaps and Detergents

- Introduction
- Raw materials
- Manufacture of soap
- Classification of detergents
- Finishing of detergents

UNIT II – Water Technology

Water

- Sources of water and constraints
- Impurities in water:
 - Dissolved
 - Suspended
 - Colloidal
- Hardness of water
- Water softening methods:
 - Lime–soda process
 - Ion-exchange process

Desalination

- Classification of desalination processes
- Evaporative processes:
 - Multi-effect evaporation
 - Multi-stage flash
 - Vapour compression
- Membrane processes:
 - Reverse osmosis
 - Electrodialysis

UNIT III – Pulp, Paper, and Sugar

Pulp and Paper

- Introduction
- Raw materials
- Types of pulp
- Manufacture of paper

Sugar

- Introduction
- Sugar extraction process
- Defecation
- Sulphitation
- Carbonation
- Concentration
- Crystallization
- Drying and refining

- Uses of molasses and bagasse
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UNIT IV – Polymers and Petroleum Refining

Polymers

- Introduction
- Degree of polymerisation
- Classification of polymers
- Polyethylene
- Polyesters

Petroleum Refining

- Introduction
 - Composition of crude oil
 - Typical refinery operations:
 - Thermal cracking
 - Catalytic cracking
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Books Recommended

1. Shreve, R.N. & Brink, J.A., *Chemical Process Industries*, 5th Edition, McGraw-Hill
 2. Austin, G.T., *Shreve's Chemical Process Industries*, 5th Edition, McGraw-Hill, 1984
 3. Dryden, C.E., Rao, M.G., *Outlines of Chemical Technology*, 3rd Edition, Affiliated East-West Press Pvt. Ltd., New Delhi, 2008
 4. Pandey, G.N., *Chemical Technology*, Volume-I, Lion Press, Kanpur
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Course Title: STRENGTH OF MATERIALS

Course Code: ESC 104

Unit I – Stresses, Strains, and Beam Loading

Stresses and Strains

- Concept of simple stress and strain
- Types of loads
- Tensile, compressive, and shear stress
- Complementary shear stress
- Thermal stresses
- Tensile test and stress–strain curve
- Hooke’s law
- Modulus of elasticity and rigidity
- Poisson’s ratio
- Principle of St. Venant strain
- Factor of safety
- Principal planes and principal stresses
- Mohr’s circle of stress
- Volumetric strain
- Elastic constants and their relations

Shearing Force and Bending Moment in Beams

- Shearing force and bending moment
 - Types of beams
 - Types of loads on beams
 - Types of supports
 - Sign conventions
 - Point of inflection
 - SFD and BMD for:
 - Concentrated loads
 - Uniformly distributed loads
 - Numerical problems
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Unit II – Bending and Deflection of Beams

Bending and Shear Stresses in Beams

- Pure bending
- Bending stress
- Composite beams
- Reinforced concrete beams
- Shear stress distribution in:
 - Rectangular sections
 - Circular sections
- Numerical problems

Deflection of Beams

- Introduction

- Macaulay's integration method
 - Simply supported beam with:
 - Central load
 - Eccentric load
 - Moment area method
 - Deflection due to shear
 - Numerical problems
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Unit III – Torsion, Columns, and Thin Shells

Torsion of Shafts

- Torsion of thin circular shafts
- Composite shafts
- Combined bending and torsion
- Equivalent torque
- Equivalent bending moment
- Numerical problems

Struts and Columns

- Definition of strut and column
- Euler's column theory and assumptions
- Columns with:
 - Both ends pinned
 - One end fixed and one end free
 - Both ends free
- Slenderness ratio
- Limitations of Euler theory
- Rankine's empirical formula
- Numerical problems

Stresses and Strains in Thin Shells

- Thin cylindrical shell under internal pressure
 - Thin spherical shell under internal pressure
 - Volumetric strain
 - Built-up shells and modifications
 - Numerical problems
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Unit IV – Springs, Strain Energy, and Failure Theories

Stresses and Strains in Springs

- Types of springs
- Close-coiled helical springs
- Open-coiled helical springs
- Leaf springs
- Springs in series and parallel
- Numerical problems

Strain Energy and Theories of Elastic Failure

- Strain energy and resilience
- Strain energy due to:
 - Tension and compression
 - Suddenly applied loads
 - Impact loads
 - Shear
 - Bending
 - Torsion
- Theories of elastic failure:
 - Maximum principal stress theory
 - Maximum shear stress theory
 - Maximum strain energy theory
- Graphical representation of failure theories
- Numerical problems

Books Recommended

1. Ryder, G.H., *Strength of Materials*, 3rd Edition (SI Units), Macmillan, 1969
2. Bedi, D.S., *Strength of Materials*, 6th Edition, Khanna Book Publishing Co.
3. Timoshenko, S., *Strength of Materials – Part I*, 3rd Edition, CBS Publishers, 1986
4. Singal & Sharma, *Strength of Materials*, Modern Publishers

Course Title: PRODUCTION AND OPERATION MANAGEMENT

Course Code: MBA 105

UNIT I – Introduction, Product & Process Design, Facility Planning

(11 Hours)

Introduction to Operations Management

- Overview of Operations Management
- Operations Management Strategy Framework
- Similarities and differences among **products, goods, and services**
- Historical evolution of operations management: changes and challenges

Product Design and Development

- Requirements and characteristics of good product design
- Product development process (technical)
- Product development techniques

Process Selection

- Types of production systems:
 - Project
 - Job
 - Batch
 - Mass
 - Process
- Product–process mix
- Design of process flows

Facility Location

- Importance of location decisions
- Location planning process
- Factors in location analysis
- Location analysis techniques

Facility Layout

- Introduction and objectives of layout
- Types of layouts
- Importance of layout decisions
- Nature of layout problems
- Assembly line balancing
- Material handling

UNIT II – Capacity, Planning, Productivity and Maintenance

(12 Hours)

Capacity Planning

- Concepts of capacity
- Factors affecting capacity
- Capacity planning process
- Capacity planning decisions

Forecasting in Production

- Demand analysis

Production Planning and Control (PPC)

- Concepts, objectives, functions
- Aggregate planning:
 - Advantages
 - Steps in developing aggregate plans
 - Aggregate planning strategies
- Scheduling:
 - Policies
 - Types of scheduling
 - Forward and backward scheduling
 - Gantt charts
 - Flow shop scheduling:
 - n jobs and 2 machines
 - n jobs and 3 machines
 - Job shop scheduling:
 - 2 jobs and n machines
 - Line of balance

Productivity Improvement Techniques

- Work study
- Method study
- Work measurement:
 - Stopwatch time study
 - Work sampling

Maintenance

- Maintenance policies for facilities and equipment
- Time to failure
- Preventive vs breakdown maintenance
- Maintenance procedures
- Total Productive Maintenance (TPM)

UNIT III – Quality Management and Six Sigma

(12 Hours)

Quality Management

- Meaning of quality and quality management
- Quality assurance
- Total Quality Management (TQM) model
- Service quality
- Just-in-Time (JIT)
- Kanban system
- Quality characteristics of goods and services

Acceptance Sampling

- Sampling by attributes
- Single and double sampling plans
- Inspection by samples
- AQL, LTPD
- Consumer's and producer's risk
- Operating characteristic (OC) curves
- Use of standard sampling tables
- Sampling by variables
- Continuous sampling plans
- Vendor rating

Tools and Techniques for Quality Improvement

- Check sheet
- Histogram
- Scatter diagram
- Cause-and-effect diagram
- Pareto chart
- Process diagram
- Statistical process control charts

Process Capability

- Meaning and significance
- Measurement and analysis
- Control charts for:
 - Variables
 - Attributes
- Establishing and interpreting control charts

Six Sigma and Advanced Quality Concepts

- Concept and application of Six Sigma
- Seven magnificent quality tools
- Introduction to experimental methods
- Taguchi method
- Quality Function Deployment (QFD):

- Concept
 - Process
 - Building the House of Quality
 - Failure Mode and Effects Analysis (FMEA):
 - Stages
 - Design
 - Process
 - Documentation
 - Benchmarking
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UNIT IV – Materials and Inventory Management, Value Engineering

(10 Hours)

Materials Management

- Concepts and objectives

Purchasing Management

- Objectives
- Functions
- Methods
- Procedures

Stores Management

- Types of stores
- Functions
- Coding methods

Inventory Management

- Concepts and classification
- Objectives
- Factors affecting inventory control policy
- Inventory costs
- Economic Order Quantity (EOQ) – introduction
- Re-order level
- ABC analysis

Value Engineering

- Introduction to value engineering/value analysis
- Objectives

- Types of values
 - Function and cost
 - Product life cycle
 - Steps and methodology in value engineering
 - FAST diagram
 - Matrix method
 - Relevance of value engineering in modern manufacturing
 - Virtual factory concept
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Text Books

1. Nair, *Production & Operations Management*, Tata McGraw-Hill
 2. Adams & Ebert, *Production & Operations Management*, Prentice Hall India
 3. Krajewski & Ritzman, *Operations Management*, Pearson Education Asia
 4. Buffa & Sarin, *Modern Production/Operations Management*, John Wiley
 5. S.N. Chary, *Production & Operations Management*, Tata McGraw-Hill
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Reference Book

1. Krajewski, L.J., Malhotra, M.K., Ritzman, L.P., Srivastava, S.K.,
Operations Management: Process and Supply Chains, 11th Edition, Pearson
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Course Title: MANAGERIAL ECONOMICS

Course Code: MBA 106

UNIT I – Introduction and Demand Analysis

(11 Hours)

Introduction to Managerial Economics

- Nature, scope, and importance of managerial economics
- The firm and its goals
- Profit maximization vs wealth maximization

- Five fundamental economic questions
- Opportunity cost
- Incremental principle
- Time perspective
- Discounting principle
- Equi-marginal principle

Demand Concepts and Analysis

- Individual and market demand
- Kinds of demand
- Determinants of demand
- Demand functions
- Demand schedule
- Law of demand

UNIT II – Consumer Behavior, Elasticity, and Forecasting

(11 Hours)

Theory of Consumer Behavior

- Cardinal utility approach
- Ordinal utility approach (indifference curve analysis)
- Derivation of demand curve using utility analysis
- Validity of utility analysis in modern times

Elasticity of Demand

- Concept and types
- Measurement of elasticity
- Importance of elasticity of demand

Demand Forecasting

- Sources of data:
 - Expert opinions
 - Surveys
 - Market experiments
 - Time series analysis:
 - Trend projection
 - Barometric forecasting:
 - Leading indicators
 - Composite indices
 - Diffusion indices
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UNIT III – Production, Cost, and Revenue Analysis

(7 Hours)

Production Function

- Concept and types
- Returns to factor
- Returns to scale
- Law of variable proportions

Cost Concepts and Analysis

- Concept of cost
- Short-run and long-run cost curves
- Relationship among cost curves
- Learning curve
- Economies of scale and economies of scope
- Estimation of cost
- Break-even analysis
- Combining break-even analysis with demand
- Degree of operating leverage
- Uses and limitations of break-even analysis

Revenue Curves

- Concept
 - Types of revenue curves
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UNIT IV – Market Structures and Pricing

(16 Hours)

Perfect Competition

- Characteristics
- Equilibrium price
- Profit-maximizing output in:
 - Short run
 - Long run

Monopoly

- Characteristics
- Equilibrium price
- Profit-maximizing output in:

- Short run
- Long run
- Price discrimination

Imperfect Competition

- Monopolistic competition:
 - Characteristics
 - Price equilibrium
- Oligopoly:
 - Types
 - Price equilibrium
 - Barriers to entry

Text Books

1. H.L. Ahuja, *Managerial Economics*, S. Chand Publishers (Latest Edition)
2. Craig Peterson, Lewis & Jain, *Managerial Economics*, Pearson Education, 1st Edition
3. Mark Hirschey, *Managerial Economics*, Thomson, 11th Edition
4. Dr. V. Panduranga Rao, *Microeconomics*, IBS Publications, 8th Edition
5. Keat, *Economic Tools for Today's Decision Makers*, 4th Edition, Pearson Education
6. William Samuelson & Stephen G. Marks, *Managerial Economics*, 5th & 6th Editions, John Wiley & Sons

Reference Books

1. Truett & Truett, *Managerial Economics Analysis: Problem Cases*, 8th Edition, Wiley
2. K.K. Dewett, *Modern Economic Theory*, S. Chand & Sons, New Delhi, 14th Edition
3. Mote, Paul & Gupta, *Managerial Economics*, Vikas Publishers, New Delhi
4. A. Koutsoyiannis, *Modern Microeconomics*, McMillan, New Delhi, 2nd Edition
5. R.L. Varshney & K.L. Maheshwari, *Managerial Economics*, S. Chand & Sons, 4th Edition

Paper Title: HEAT TRANSFER LAB

Paper Code: PCC153

Credits: 1.5

Maximum Marks: 50

List of Experiments

1. Determination of heat transfer coefficient for different types of heat transfer equipment. **Wilson plots.**
 2. Unsteady state heat transfer in jacketed vessels (**Open pan evaporator**).
 3. Correlation of instantaneous heat transfer coefficients with time study deposition of scale on a heating surface.
 4. Determination of heat losses for insulated pipes.
 5. Study of double pipe heat exchanger and determination of overall heat transfer coefficient.
 6. Study the performance characteristics of a **1–2 shell and tube heat exchanger**.
 7. Study and operation of **long tube, forced circulation and multiple effect evaporators**.
 8. **Duhring plot** for solutions involving non-volatile solutes.
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Chemical Technology (Organic) Lab

Paper Code: PCC 156

Marks: 50

Credits: 1.5

Course Outcomes

- Determination of **Acid value**
- Determination of **Iodine value**
- Determination of **Saponification value**
- Determination of **Reducing and Non-reducing sugars** by:
 - (i) Fehling's method
 - (ii) Pavy's method
- Determination of **Free and combined alkali, Total fatty matter, and Moisture**

List of Practicals

1. **Oils and Fats:** Determination of acid value, iodine value, and saponification value.
 2. **Carbohydrates:** Determination of reducing and non-reducing sugars by:
 - (i) Fehling's method
 - (ii) Pavy's method
 3. **Soaps:** Determination of free and combined alkali, total fatty matter, moisture, and insoluble matter.
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Title: COMPREHENSIVE VIVA

Code: CHE 101

Credits: 1

L T P: – – –

Maximum Marks:

- End Term: 50
- Mid Term: –
- Practical: –

Elective: No

Description

The viva-voce examination will be **comprehensive** and shall cover **all subjects taught during the first to fourth semesters**.

Title: PROCESS EQUIPMENT DESIGN

Code: ESC 155

Credits: 1.5

L T P: – – 3

Maximum Marks (Practical): 50

Elective: No

Contact Hours: 45

Objectives

- To be familiar with the **process and mechanical aspects** of design of process equipment.
- To understand **design factors, procedures, codes, and standards**.

Course Outcomes

Students should be able to apply the criteria involved in the design of:

- Pressure vessels
- Vessel supports
- Various types of joints
- Foundations
as per relevant **codes and specifications**.

List of Practicals

1. Study of factors influencing vessel design; classification of pressure vessels, applications, methods of fabrication, fundamental principles, and equations.
2. Study of pressure vessel codes, specifications, and standards; review of code development, ASME codes, API-ASME code, Section VIII of ASME codes.
3. General design considerations: design pressure, design temperature, materials, design stress (nominal design strength), corrosion allowance, design loads, and minimum practical wall thickness.

4. Design of thin-walled vessels under internal pressure: cylindrical and spherical shells, heads and closures, flat ends, domed ends, conical sections, and end closures.
5. Design of vessels under external pressure: cylindrical shells, stiffening rings, and vessel heads.
6. Design of vessels under combined loading: weight loads, wind loads (tall vessels), and torque.
7. Design of welded joints and bolted flanged joints.
8. Design of foundations and supports.

Books Recommended

1. Bhattacharyya, B.C. – *Introduction to Chemical Equipment Design (Mechanical Aspects)*, CEED.
 2. Brownell and Young – *Process Equipment Design*, Wiley Publication.
 3. Joshi, M.V. – *Process Equipment Design*, Macmillan India.
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